

In other pages, we obtained one of the greatest results, the Primary decomposition.

Now, we will walk into deeper result, the Second decomposition theorem. in the right hand is the Primary decomposition, in the left hand is the observation for cyclic subspace.

The Primary decomposition theorem.

Let T be a linear operator on the finite-dimensional vector space V over a field F . Let P be the minimal polynomial for T with the irreducible decomposition

$$P = p_1^{r_1} \cdot p_2^{r_2} \cdots p_k^{r_k}$$

Then,

$$V = \bigoplus_{i=1}^k \ker p_i^{r_i}(T) \text{ and each } \ker p_i^{r_i}(T) \text{ is invariant } T.$$

Moreover, the restriction of T by $\ker p_i^{r_i}(T)$ has the minimal polynomial as $p_i^{r_i}$.

Thm. Let α be any non-zero vector in V and let P_α be the T -annihilator of α .

- 1) $\deg P_\alpha = \dim Z(\alpha, T)$
- 2) $Z(\alpha, T) = \text{span}\{T^k \alpha \mid 0 \leq k \leq \deg P_\alpha\}$
- 3) The minimal polynomial of $T|_{Z(\alpha, T)}$ is P_α .

Cyclic decomposition Theorem.

Thm. Let $T: V \rightarrow V$ be a linear operator on finite-dimensional space V .

Suppose that the minimal polynomial of T has a irreducible decomposition

$$m_T(x) = p_1(x)^{r_1} \dots p_k(x)^{r_k}$$

Then, by the primary decomposition, V is decomposed by T -invariant spaces

$$V = \ker p_1^{r_1} \oplus \dots \oplus \ker p_k^{r_k}$$

With $T|_{\ker p_i^{r_i}}$ has minimal polynomial as $p_i^{r_i}$ 여기서 r_i 가 Z 의 블록의 수.

Now, for fixed $1 \leq i \leq k$, denote $T_i = T|_{\ker p_i^{r_i}}$ and $W_i = \ker p_i^{r_i}$.

Then, there are non-zero vectors $\alpha_1, \dots, \alpha_n$ such that

$$\ker p_i^{r_i} = Z(\alpha_1, T_i) \oplus \dots \oplus Z(\alpha_n, T_i)$$

satisfying for some unique integers $1 \leq r_n \leq \dots \leq r_2 \leq r_1 = r_i$,

$$m_{T_i|Z(\alpha_s, T_i)} = p_i^{r_s}$$

Further, $\dim Z(\alpha_s, T_i) = \deg p_i^{r_s}$.

Computation.

Let A be a 8×8 Matrix over $\mathbb{Q}$ as

$$A = \begin{bmatrix} 1 & 1 & & & & & & \\ 0 & 1 & & & & & & \\ & & 1 & 0 & 0 & & & \\ & & 0 & 0 & -1 & & & \\ & & 0 & 1 & 0 & & & \\ & & & & & 0 & 0 & 2 \\ & & & & & 1 & 0 & 0 \\ & & & & & 0 & 1 & 0 \end{bmatrix} \quad \text{Denote } A_1 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix},$$

$$A_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$A_3 = \begin{bmatrix} 0 & 0 & 2 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad \text{Find the Rational form.}$$

1. The characteristic Polynomial is

$$[(t-1)^2-0] \cdot [(t-1) \cdot (t^2+1)] \cdot [t^3-2] = (t-1)^3 \cdot (t^2+1) \cdot (t^3-2).$$

2. The minimal polynomial is: Since

$$A^2 = \begin{bmatrix} A_1 & & \\ & A_2 & \\ & & A_3 \end{bmatrix}^2 = \begin{bmatrix} A_1^2 & & \\ & A_2^2 & \\ & & A_3^2 \end{bmatrix}.$$

So the minimal polynomial of A is the least common multiple,

$$m_A = \text{lcm}(m_{A_1}, m_{A_2}, m_{A_3}) = (t-1)^2 \cdot (t^2+1) \cdot (t^3-2).$$

By the primary decomposition theorem,

$$V = \ker(A-I)^2 \oplus \ker(A^2+I) \oplus \ker(A^3-2I)$$

Meanwhile, $\dim \ker(A-I)^2 = \deg \chi_{A_1} = 3$, and the calculation gives

$$\ker(A-I)^2 = \text{Span}\{e_1, e_2, e_3\} \text{ and } Ae_2 = e_1 + e_2, \text{ so } Z(e_2, A) = \text{Span}\{e_2, Ae_2\} = \text{Span}\{e_1, e_2\}.$$

and, $Ae_3 = e_3$, so $Z(e_3, A) = \text{Span}\{e_3\}$, the annihilator $\text{Ann}(e_3, A) = t-1$.

Therefore, $\ker(A-I)^2 = Z(e_2, A) \oplus Z(e_3, A)$

with the minimal polynomial $(t-1)^2$, $t-1$.

Debris

$$A^2 + I = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 2 & 0 \\ 0 & 0 & 2 \\ 1 & 0 & 0 \end{bmatrix}$$

$$A^3 - 2I = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

first, observe that

$$= \begin{bmatrix} 2 & 2 & & & & & & \\ & 0 & 2 & & & & & \\ & & & t_2 & & & & \\ & & & & 0 & & & \\ & & & & & 0 & & \\ & & & & & & t_1 & \\ & & & & & & & 2 & 0 \\ & & & & & & & & 0 & 1 & 2 \\ & & & & & & & & & & 1 & 0 & 1 \end{bmatrix}$$

$$+ I_8$$

$$- 2I_8 \quad \begin{bmatrix} -1 & 3 & & & & & & \\ & 0 & -1 & & & & & \\ & & & -1 & 0 & 0 & & \\ & & & 0 & -2 & 1 & & \\ & & & 0 & -1 & -2 & & \\ & & & & & & 0 & \\ & & & & & & & 0 & \\ & & & & & & & & 0 \end{bmatrix}$$

Similarly, $\ker(A^2 + I) = \text{span}\{e_4, e_5\} = Z(e_4, A)$, since $Ae_4 = e_5$.
 $\ker(A^3 - 2I) = \text{span}\{e_6, e_7, e_8\} = Z(e_6, A)$, since $Ae_6 = e_7$, $A^2e_6 = e_8$.

Consequently

$$V = Z(e_3, A) \oplus Z(e_2, A) \oplus Z(e_4, A) \oplus Z(e_6, A)$$

With $(t-1)$ $(t-1)^2$ t^2+1 t^3-2 .
 Now, the rational form is:

$$[A]_C = \begin{bmatrix} 1 & & & & & & & 2 \\ & t-1 & & & & & & -4 \\ & & 1 & 0 & & & & 4 \\ & & & 1 & 0 & & & -5 \\ & & & & 1 & 0 & & 4 \\ & & & & & 1 & 0 & -2 \\ & & & & & & 1 & 2 \end{bmatrix}$$

$$A \sim C$$

$$\Leftrightarrow B^{-1}AB = C$$

Theoretical Exercises in HK.

7.2.2. Let T be a linear operator on the fin. dim V . Then,

$\ker T$ and $\operatorname{Im} T$ are independent if and only if

$\ker T$ is the unique T -invariant Complementary of $\operatorname{Im} T$.

Proof. Suppose that $\ker T$ and $\operatorname{Im} T$ are independent.

Given $\alpha \in \ker T \cap \operatorname{Im} T$, $\alpha + (-\alpha) = 0$ and $-\alpha \in \ker T \cap \operatorname{Im} T$,

thus the independency gives $\alpha = 0$. Therefore, $\ker T \cap \operatorname{Im} T = \{0\}$. General Property of independence.

Let $\{\alpha_1, \alpha_2, \dots, \alpha_k\}$ be a basis for $\operatorname{Im} T$.

And, by the dimension theorem, the number of elements of $\ker T$ is $n-k$,

so let $\{\alpha_{k+1}, \dots, \alpha_n\}$ be the basis for $\ker T$. Dimension theorem & reduction

Again, the independency gives that $\{\alpha_1, \dots, \alpha_k, \alpha_{k+1}, \dots, \alpha_n\}$ are linearly independent, so this is basis for V , which proves that

$$V = \ker T \oplus \operatorname{Im} T.$$

The T -invariantness is obtained by: $T(\alpha) = 0 \Rightarrow T(T(\alpha)) = T(0) = 0$.

To show the uniqueness, let $W < V$ satisfies $V = W \oplus \operatorname{Im} T$ and $T[W] \subseteq W$.

Given $\alpha \in W$, $T(\alpha) \in W \cap \operatorname{Im} T = \{0\}$, so $\alpha \in \ker T$. 0 is independent & 1/0 is X.

Therefore, $W \subseteq \ker T$ and since $\dim W = \dim V - \dim \operatorname{Im} T = \dim \ker T$,

$$\text{so } W = \ker T.$$

The Converse is fucking clear.

7.2.5. Let T be a linear operator on the vector space V over F . Suppose $V_1, \dots, V_k \leq V$ are T -invariant subspaces and $V = V_1 \oplus V_2 \oplus \dots \oplus V_k$. Given a polynomial $f \in F[t]$,

$$f(T)[V] = f(T)[V_1] \oplus \dots \oplus f(T)[V_k].$$

Moreover, if the T -annihilator of α and β are same, then the T -annihilator of $f(T)(\alpha)$ and $f(T)(\beta)$ are same.

Proof. Given $\alpha \in V$, α has a unique representation

$$\alpha = \alpha_1 + \dots + \alpha_k \text{ where } \alpha_i \in V_i.$$

Then,
$$f(T)(\alpha) = f(T)(\alpha_1) + \dots + f(T)(\alpha_k)$$

and each $f(T)(\alpha_i) \in V_i$, since each V_i is T -invariant.

And, suppose that for some $f(T)(\beta_i) \in f(T)[V_i]$ ($1 \leq i \leq k$),

$$f(T)(\alpha_1) + \dots + f(T)(\alpha_k) = f(T)(\beta_1) + \dots + f(T)(\beta_k).$$

Then

$$f(T)(\alpha_1 - \beta_1) + \dots + f(T)(\alpha_k - \beta_k) = 0$$

and $\alpha_i - \beta_i \in V_i$, so $f(T)(\alpha_i - \beta_i) \in V_i$, so

$$f(T)(\alpha_1 - \beta_1) = \dots = f(T)(\alpha_k - \beta_k) = 0,$$

It proves the representation of $f(T)(\alpha)$ is unique. Consequently

$$f(T)[V] = f(T)[V_1] \oplus \dots \oplus f(T)[V_k]. \text{ It proves the first.}$$

Now, let $p(t) \in F[t]$ be the T -annihilator of α and β .

That is, $g(T)(\alpha) = 0$ and $h(T)(\alpha) = 0$ implies $p(t) \mid g(t), h(t)$.

Now, suppose that $r(t) \in F[t]$ satisfies that $r(T)(f(T)(\alpha)) = 0$.

Then, $r(T)f(T)(\alpha) = f(T)r(T)(\alpha) = 0$, so that $p(t) \mid f(t) \cdot r(t)$.

7.2.11.

Prop. Let A and B be 3 by 3 matrices over a field F . Then, A is similar to B if and only if $\chi_A = \chi_B$ and $m_A = m_B$

where χ and m are the characteristic and the minimal, respectively.

Proof. For a cubic polynomial f , there are only three possible cases:

f is irreducible or f has a linear factor and degree 2 irreducible factor or f splits over F into three linear factors.

Assume $\chi_A = \chi_B$ and $m_A = m_B$. Simply, denote $\chi = \chi_A$ and $m = m_B$.

Suppose that χ is irreducible. Then, since the minimal polynomial divides the characteristic, so $\chi = m$. Now, A and B both are similar to the matrix

$$C = \begin{bmatrix} 0 & 0 & -c_0 \\ 1 & 0 & -c_1 \\ 0 & 1 & -c_2 \end{bmatrix} \text{ where } \chi(t) = t^3 + c_2 t^2 + c_1 t + c_0.$$

Suppose that χ factors into a linear factor and a degree 2 irreducible factor. But, since the minimal polynomial divides the characteristic, m is either the linear or the degree 2 factor or f itself.

But, the minimal polynomial is linear implies that T is just multiple of identity, so the characteristic cannot have irreducible of degree 2, so it cannot happen.

And, if the minimal polynomial is degree 2 irreducible, but since m and χ shares the roots, so m has a root in the given field, so it cannot be irreducible degree 2. Therefore, $m = \chi$, which implies that $A \sim C$ and $B \sim C$ where C is a matrix above, so $A \sim B$.

Suppose that χ factors into three linear factors, as $\chi = (t-a)(t-b)(t-c)$.

If the minimal has degree 1, WLOG, $m = t-a$, then $A = B = aI$.

If the minimal has degree 2, WLOG, $m = (t-a)(t-b)$, then $C = a$ or b , and

$$\text{then } A, B \sim \begin{bmatrix} c & & \\ 0 & -ab & \\ 1 & a+b & \end{bmatrix}$$

Finally, if $m = \chi$, then again $A, B \sim C$. It completes the classification.

Computational Exercise.

Let $A \in M_{13 \times 13}(\mathbb{R})$ be a matrix satisfying

$$\chi_A(t) = (t^2+1)^5 \cdot (t-2)^3 \quad \text{and} \quad m_A(t) = (t^2+1)^3 \cdot (t-2)^2.$$

with $\dim \ker(A^2+I) = 4$, $\dim \ker(A^2+I)^2 = 8$, $\dim \ker(A^2+I)^3 = 10$,
 $\dim \ker(A-2I) = 2$, $\dim \ker(A-2I)^2 = 3$.

Find the rational canonical form of A .

Proof. By the Primary decomposition,

$$\mathbb{R}^{13} = \ker(A^2+I)^3 \oplus \ker(A-2I)^2.$$

And, observe that the dimensions, we obtain

$$\ker(A^2+I)^3 = \underset{4}{W_1} \oplus \underset{6}{W_2} \quad \text{where} \quad m_{A|W_1} = (t^2+1)^2, \quad m_{A|W_2} = (t^2+1)^3,$$

and $\ker(A-2I)^2 = \underset{1}{U_1} \oplus \underset{2}{U_2}$ where $m_{A|U_1} = (t-2)$, $m_{A|U_2} = (t-2)^2$.

Consequently, the invariant factors are

$$f_2 = (t^2+1)^2(t-2), \quad f_1 = (t^2+1)^3 \cdot (t-2)^2$$

and so that the rational canonical form is

$$R_A = C(f_2) \oplus C(f_1)$$

where $C(f)$ is the Companion Matrix for f . $\square$